

Xingjian Ren

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Summary

Senior at Georgia Institute of Technology majoring in Mathematics and Computer Science, with strong foundations in algorithms, software development, and applied math. Experienced in building full-stack web applications, designing interactive game systems, and implementing machine learning pipelines. Proficient in Java, Python, C++, and JavaScript. Passionate about solving problems with logic and programming and applying these skills into real-world software engineering internships.

Skills

- **Programming Languages:** Java, Python, C++, C, C#, SQL, JavaScript, HTML/CSS.
- **Frameworks & Libraries:** React, Django, PyTorch, scikit-learn, TensorFlow, NumPy, Pandas, Streamlit.
- **Developer Tools:** Git, Docker, VS Code, PyCharm, IntelliJ, Unity, Jira
- **Core Competencies:** Data Structures & Algorithms, System Design, Object-Oriented Programming (OOP), Agile (Scrum), Full-Stack Development, UI/UX Design.

EDUCATION

Georgia Institute of Technology, College of Sciences August 2023 – (Expected) May 2027
Concentrated in **Data Science** (Math); **Information Internetworks** and **Intelligence** (Computer Science), GPA: 3.89

Experience

Teaching Assistant — CS 2110: Computer Organization & Programming, Georgia Tech August 2025 – December 2025

- Helped 500+ students to learn logic circuits, assembly language, and C programming with the TA team.
- Taught lab sections for 58 students, held office hours, answered questions on Piazza, designed homework, and graded homework and quizzes.
- Debugged codes for multiple students during office hours.

Academic Projects

Group, Machine Learning Project — Python, NumPy, GitHub, PyTorch, Scikit-learn September 2025 – December 2025

- Built an end-to-end machine learning pipeline for brain tumor MRI classification.
- Implemented the preprocessing pipeline using OpenCV to preprocess brain tumor MRI data for the models.
- Trained and compared multiple supervised and unsupervised models.
- Evaluated models using accuracy, confusion matrices, precision/recall/F1, and confidence analysis.
- Conducted comparative analysis between the models' performances using the evaluation data.
- Developed a GitHub repository and GitHub Pages site with a report including introduction, data preprocessing, model selections, results, and the analysis with data provided.

Current Crisis VIP: Research Tree Team (In Progress) — C#, Unity, GitHub, VS Code August 2025 – Present

- Designing and implementing the research tree for a power grid resilience simulation game.
- Developing research nodes that dynamically influence game economy and player outcomes, including power plant costs, city revenue, and citizen happiness.
- Collaborating with an interdisciplinary team using Unity for prototyping and GitHub for version control.
- Producing design documents and a website introducing the game.
- Personally implemented 4 research nodes involving power plants upgrades in costs, efficiency and placement, with corresponding UI indicators reflecting the upgrades.

Group, Pokémon Trading Web App — Python (Django), JavaScript, HTML, SQL, Agile (Scrum) March 2025 – May 2025

- Built a full-stack application for users to catch and trade Pokémon using Django and PostgreSQL.
- implementing RESTful principles for seamless data exchange between the frontend and backend.
- Developed frontend features (stat bars, filters, sorting, favorite markings) and backend functionality for collections, shopping cart, and marketplace.
- Created interactive UI components for 5,000+ Pokémon using PokeAPI with clickable image-based cards.
- Implemented backend trading logic: user balance verification, inventory updates, duplicate-prevention checks.
- Integrated authentication and access control, including admin-only and login-required views.
- Collaborated under Agile Methodology and via GitHub to manage version control, pull requests, and issue tracking.

Individual, Game Design — Java (JavaFX), IntelliJ September 2024 – December 2024

- Redesigned and implemented mechanics for Name Arena, Minesweeper, and Wordle.
- Applied data structures and algorithms (HashMap, PriorityQueue, Stack, ArrayList, sorting, greedy) to create efficient gameplay.
- Built desktop applications with JavaFX and IntelliJ, with interactive and humorous UIs for user-friendly mechanics.
- Conducted playtesting and iteratively refined designs based on peer feedback.